INCH-POUND
MIL-DTL-14522E
w/AMENDMENT 1
7 May 2024
SUPERSEDING
MIL-DTL-14522E
17 May 2010

DETAIL SPECIFICATION

COMPONENTS, PREFABRICATED STRONG ROOM, PORTABLE: FOR SMALL ARMS AND EQUIPMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers portable, prefabricated strong room components, for small arms and equipment. It includes all structural components, clamps, and fasteners necessary to assemble strong rooms of various sizes as illustrated in figure 1.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3 and section 4 of this specification, whether or not they are listed.

2.2 Government documents

2.2.1. Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Avenue, Philadelphia, PA 19111-5096 or email CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil .

AMSC N/A FSC 5450

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COMMERCIAL ITEM DESCRIPTIONS

A-A-1923 - Shield Expansion (Lag, Machine and Externally Threaded Wedge Bolt Anchors)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-T-704 - Treatment and Painting of Materiel

(Copies of these documents are available online at https://assist.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. ARNY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

NATICK Drawing 13218E8075 - Strong Room, Prefabricated Portable

(Copies of drawings are available from U.S. Army Natick Research, Development, and Engineering Center, ATTN: SATNC-WEE, Natick, MA 01760-5018).

2.3 Non-Government publications. The following documents form a part of this document the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASME International

ASME B18.2.1 - Square and Hex Bolts and Screw (Inch Series)
ASME SECTION IX - Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators

(Copies of these documents are available from www.asme.org or ASME International, Three Park Avenue, M/S 10E, New York, NY 10016-5990.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from www.asq.org or the American Society for Quality, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)

AMERICAN WELDING SOCIETY (AWS)

AWS A5.1/A5.1M - Specification for Carbon Steel Electrodes for Shielded Metal Arc-Welding
AWS A5.18/A5.18M - Specification for Carbon Steel Electrodes and Rods for Gas Shielded Arc Welding
AWS D1.1/D1.1M - Structural Welding Code-Steel

(Copies of these documents are available from www.aws.org or the American Welding Society, 550 N.W. LeJeune Road, Miami, FL 33126.)

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THE SOCIETY FOR PROTECTIVE COATINGS

SSPC-SP 1 - Solvent Cleaning

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AMS-STD-595 Colors Used in Government Procurement

(Copies of this document are available from www.sae.org or the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample of each component shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials and components. The materials and components shall be as specified herein and as shown on drawing 13218E8075. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided that the completed item complies with all the provisions of this document. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2.1 Welding electrodes. Welding electrodes shall conform to AWS A5.1/A5.1M or AWS A5.18/A5.18M. Unless otherwise specified, the class of electrodes used will be at the contractor's option.

3.2.2 Wire mesh. Wire mesh shall be as specified in figure 2.

3.2.3 Wall and floor clamp. When the wall and floor clamp is specified (see 6.2), it shall be furnished with a hex lag screw, 0.375 x 2.00, steel per ASME B18.2.1. When the clamp is to be applied to concrete or masonry surfaces, an expansion anchor shield conforming to A-A-1923, dash number 112 shall be furnished with each clamp.

3.3 Design. Components for prefabricated strong rooms shall be in accordance with Drawing13218E8075 (see 6.2) and as specified herein. The drawings forming a part of this document is an end product drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the strong room components. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.4 Identification marking. All strong room components shall be metal stamped as shown on drawing 13218E8075. All metal stamping shall be legible after painting.

3.5 Finish.

3.5.1 Cleaning. After fabrication, each strong room component shall be prepared for painting in accordance with SSPC-SP 1.

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3.5.2 Treatment and painting. Each component shall be treated and painted in accordance with MIL-T-704 except that, unless otherwise specified (see 6.2), the finish coat shall be semigloss black enamel, Color No. 27038 in accordance with SAE AMS-STD-595.

3.6 Workmanship.

3.6.1 Casting. All cast strong room components shall be free from harmful extraneous material such as sand, dirt, pits, sprues, scale and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.6.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material; corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing, providing all slag is removed after burning. All bends shall be made with controlled means to ensure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.6.3 Machine work. Tolerance and gages for metal fits shall conform to the limits specified on drawing 13218E8075 and to the standard of the industry.

3.6.4 Welding. The surfaces to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure and shall have no crack type defects when the component is completed. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the components to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 1/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified, the welding process used in fabrication of the component shall be at the option of the contractor.

3.6.5 Welders. Before assigning any welder to welding work covered by this document, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1/D1.1M, Structural Welding Code-Steel

SME Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators

Contractors who only make horizontal welds need not qualify welders for “all position welding”. Subject to approval by the Government, the contractor’s standard welder qualification may be substituted in lieu of the above codes provided that the contractor’s procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds. For the purpose of this document, automatic welding equipment operators shall be considered welders and subject to the qualification procedures outlined above.

4. VERIFICATION

4.1 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such Items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3. The presence of any defect shall be cause for rejection of the first article.

4.4.1 Conformance inspection. Unless otherwise specified, sampling for inspect ion shall be performed in accordance with ASQ Z1.4. Unless otherwise specified, the Acceptable Quality Limits (AQLs) listed in this section shall be used to establish the sample sizes, however, the acceptance number shall be zero.

4.4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Certification. The contractor shall furnish a certificate of compliance for the requirements specified in 3.2.1 and 3.6.5.

4.4.2 End item visual examination. Each type of component shall be examined for the applicable defects listed below. The lot size shall be expressed in units of all strong room components. The sample unit shall be one component. The percentage of samples of each component in relation to the total of all components sampled shall be proportional to the percentage of that type of component in the lot in relation to the total of all components in the lot. The inspection level shall be II and the acceptable quality limit (AQL), expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Material	Not as specified
Wire mesh	Not as specified
Wall and floor	Not as specified, when specified clamp anchor shield missing or not as specified
Components	Missing or not as specified
Identification	Not as specified markings
Finish	Cleaning, treatment, or painting not as specified
Workmanship	Not as specified

4.4.3 End item dimensional examination. Each type of component shall be examined for conformance to the dimensions annotated with an asterisk on drawing 13218E8075. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of all strong room components. The sample unit shall be one component. The percentage of samples of each component in relation to the total of all components sampled shall be proportional to the percentage of that type of component in the lot in relation to the total of all components in the lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Services or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Each component is intended for use in the construction of various sizes of strong Rooms. These strong rooms are used for the safe storage of small arms and equipment in National Guard Armories and similar installations.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article in required (see 3.1, 4.3, ad 6.3).
c. When the wall and floor clamp is required and when the anchor shield is required (see 3.2.3).
d. Components required and quantity of each component required (see 3.3).
e. Color required when other than as specified (see 3.5.2).
f. Packaging requirements (see 5 .1).

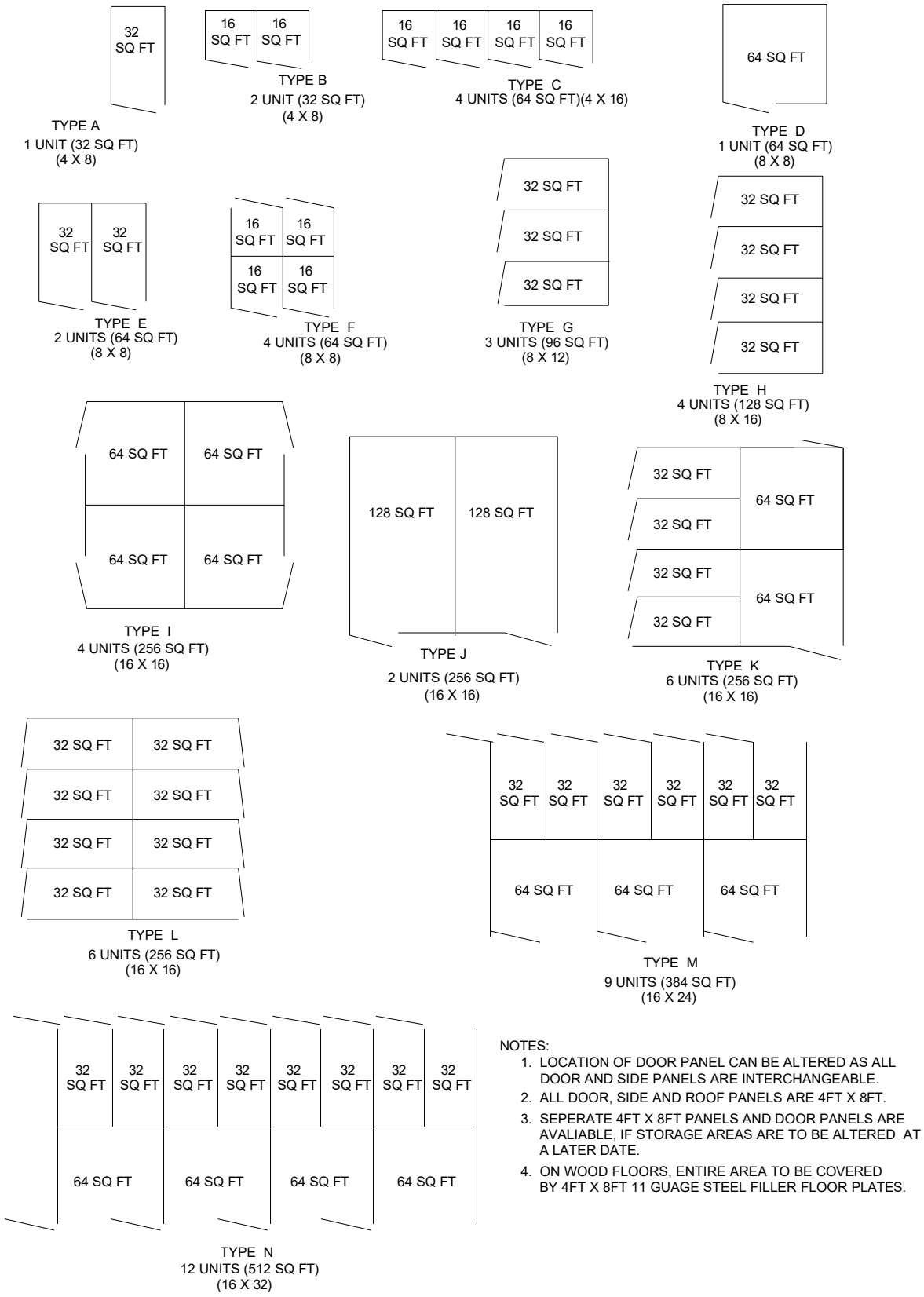
6.3 First article. When a first article is required, it will be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Assemble
Installations
Safe storage
Structural

6.5 Amendment Notation. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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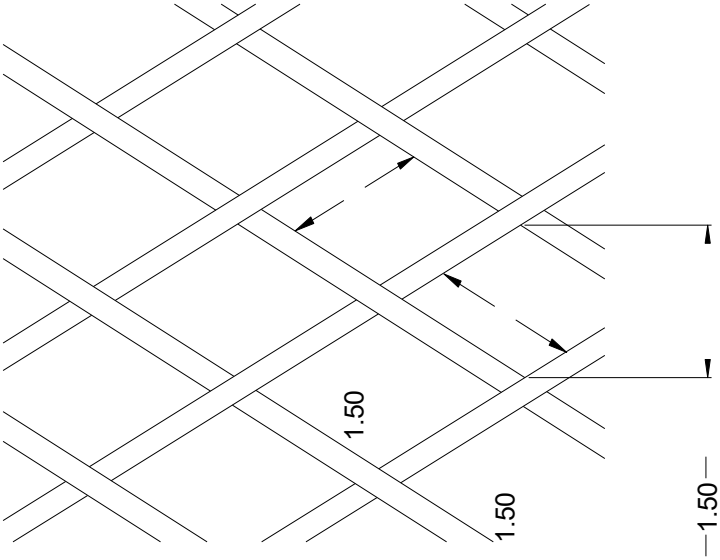


NOTES:

1. LOCATION OF DOOR PANEL CAN BE ALTERED AS ALL DOOR AND SIDE PANELS ARE INTERCHANGEABLE.
2. ALL DOOR, SIDE AND ROOF PANELS ARE 4FT X 8FT.
3. SEPERATE 4FT X 8FT PANELS AND DOOR PANELS ARE AVALIABLE, IF STORAGE AREAS ARE TO BE ALTERED AT A LATER DATE.
4. ON WOOD FLOORS, ENTIRE AREA TO BE COVERED BY 4FT X 8FT 11 GUAGE STEEL FILLER FLOOR PLATES.

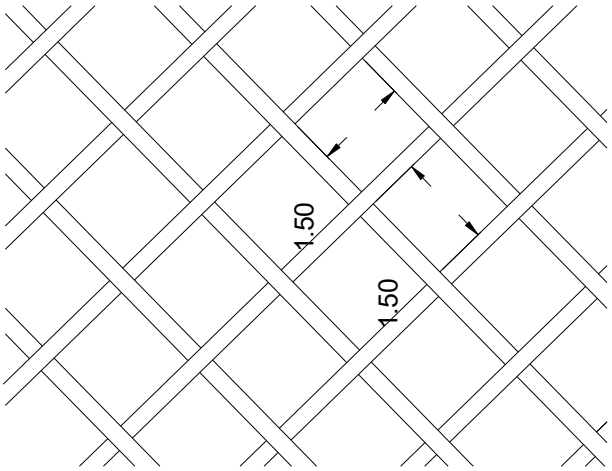
FIGURE 1. Typical strong room combinations.

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LONG DIAMOND

NOTES:
1. WIRE TO BE Ø.160-Ø.165, 1008 TO 1010 STEEL.
2. MANUFACTURERS STANDARD COMMERCIAL TOLERANCES SHALL APPLY.



SQUARE DIAMOND

FIGURE 2. Diamond wire mesh.

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Custodians:
Army - CE
Navy - AS

Review activity:
Army - AV
Navy - MC

Preparing activity:
DLA – CQ

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